

Technische Hochschule Ingolstadt
Fakultät Elektro- und Informationstechnik

Sensor Network Technologies and Sensor Fusion

Occupancy Grids

Early Fusion

Jan 14, 2025

Welcome to the lecture!

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AIMotion, AI and Sensor Data Fusion

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Sensor Networks and Sensor Fusion

Organizational Matters

- **Please register for the lecture on Moodle!**
  - Link: https://moodle.thi.de/course/view.php?id=10580
  - Key: **SensNT+DF#WS25**
- **News on Moodle:**
  - Slides for the lecture (upcoming lectures marked as WIP and are subject to change!)
  - Formulary, Exercises (Solutions provided after discussion in the lecture)
  - Exercises are optional, one task / sheet per week, but are highly recommended.
    They can be handed in for correction.
  - I will *try* to publish exercises before Wednesday‘s lecture so that initial questions can be discussed on Thursday.
- **Exam: Written exam, 90 minutes**
- **Consultation hours: Please arrange via e-mail to bjoern.giesler@thi.de.**
- **Please also don‘t hesitate to send questions, lecture feedback, change requests, etc.!**

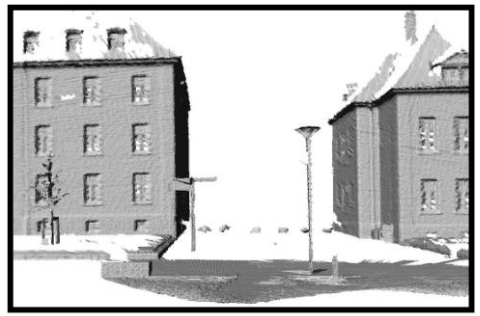
- **Previous lectures:**
  - Mathematical details of Particle Filters
  - Joint Probability Data Association Filters
  - JPDAs and Particle Filters
  - Simultaneous Localization and Mapping (based on Kalman Filters)
- **This week:**
  - Simultaneous Localization and Mapping (based on Particle Filters)
  - Overview over features for SLAM
  - Occupancy Grids and Grid-based SLAM
- **Upcoming:**
  - (DNN-based) early fusion approaches.

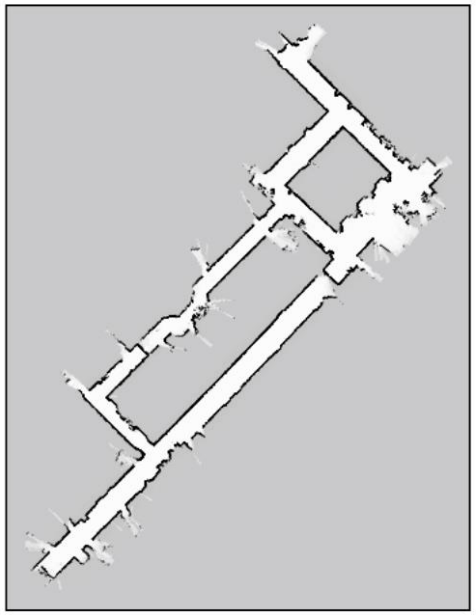
Let's talk exam!

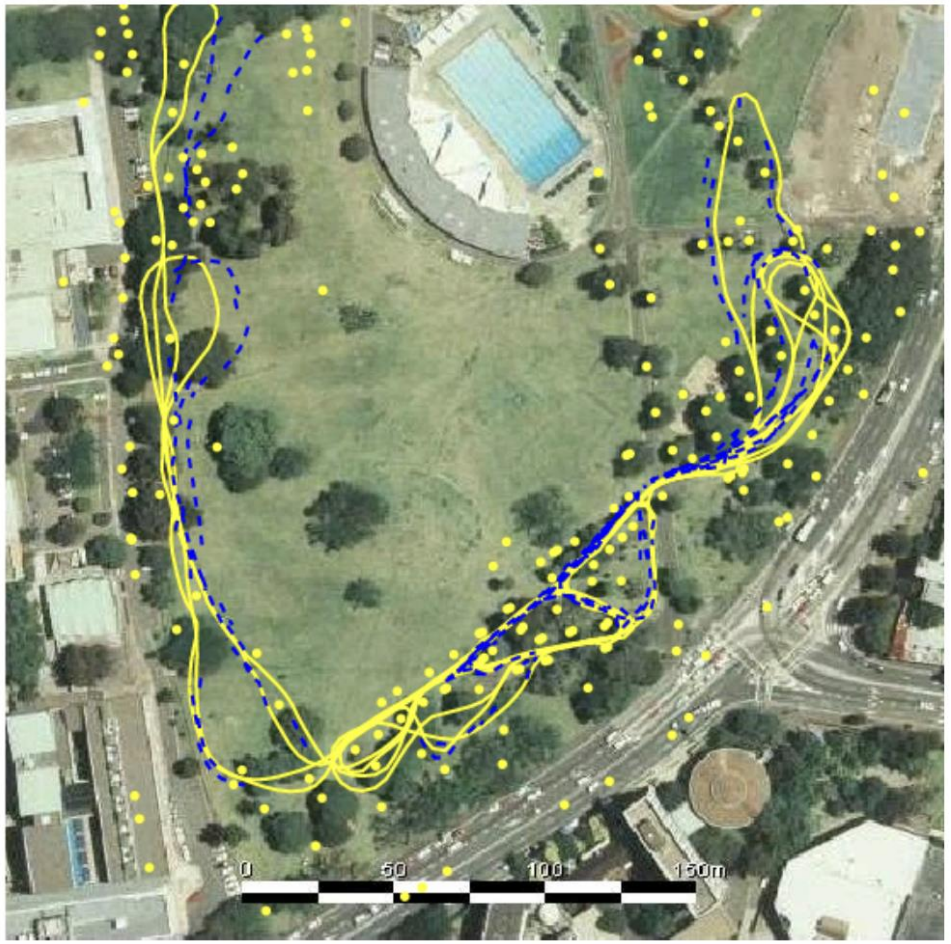
- **I want to know what's in your brain! My priority is always**
  1. have you understood the concept?
  2. ...thoroughly enough so you could implement it practically?
  3. and last, are the numerical results you obtain correct?
- **No formulary, personal notes, tablets, laptops, AI assistants or personal humanoid messenger or telepresence robots allowed.**
- **Pocket calculators are a necessity.**
- **There will likely be some hand calculation with the help of pocket calculators.**
- **No coding of course, but questions about algorithms in pseudo code are possible.**
- **Long answer questions possible. Multiple choice questions possible. Formula questions possible.**
- **Basically everything we discussed in the lecture is relevant. Do not rely on only the exercise sheets or programming exercises to prepare.**
- **I will be fair!**

Occupancy Grids and Grid-based SLAM

Feature Maps vs Volumetric Maps

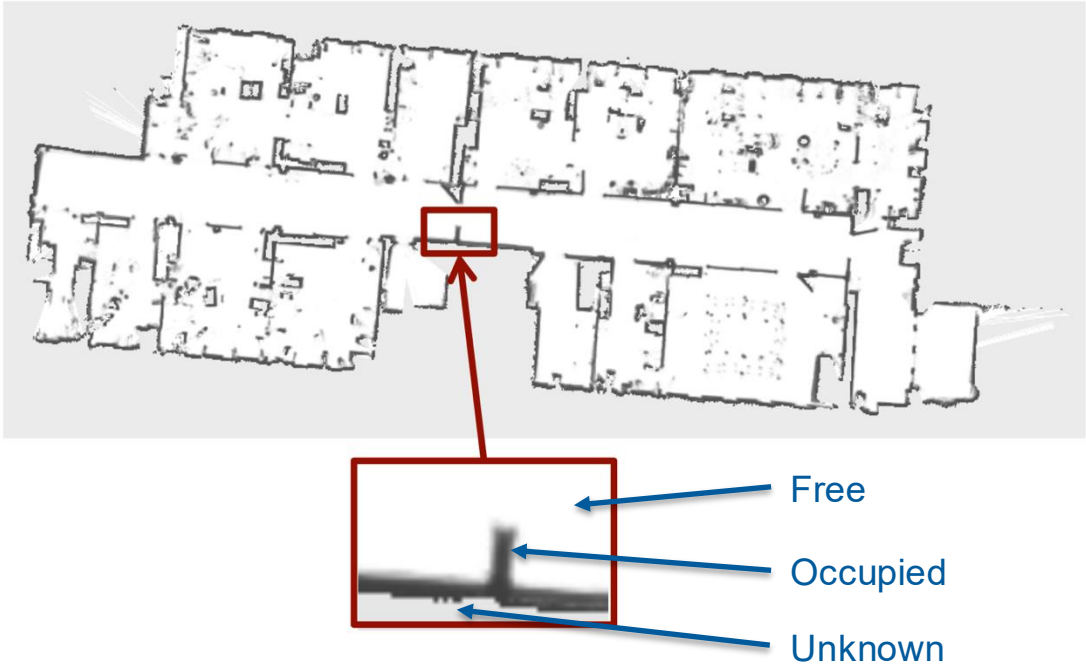






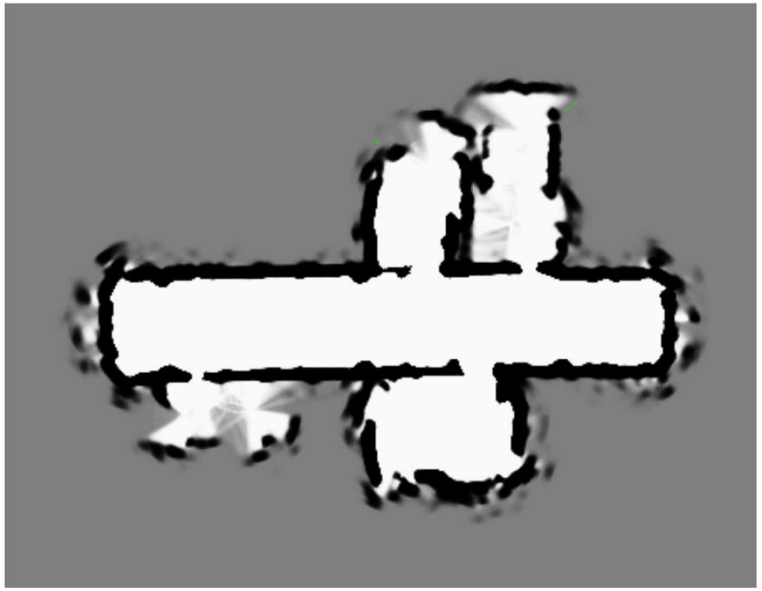
Features vs Grids

- **So far, we have only used feature maps.**
  - Natural choice for Kalman filter-based SLAM systems
  - Multiple feature observations improve the position estimate (EKF)
  - Compact representation
- **Grid maps:**
  - Discretize the world into cells
  - Grid structure is rigid
  - Each cell is assumed to be occupied or free space
  - Non-parametric model
  - Requires substantial memory resources
  - Does not rely on a feature detector



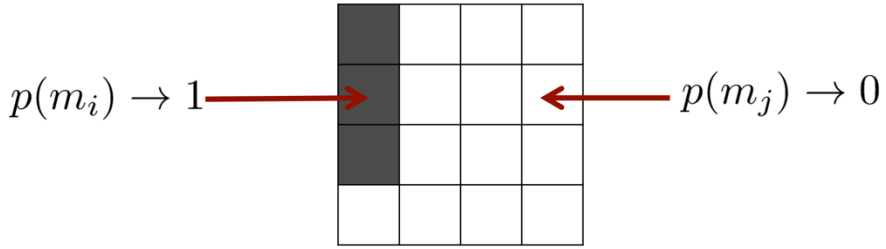
Occupancy Grid Mapping

- **Developed in the mid 80'ies by Moravec and Elfes [1]**
- **Originally developed for noisy sonars**
- **Also called "mapping with known poses"**





Assumptions



1. **The area that corresponds to a cell is either completely free or occupied**
   - We can model each cell with a binary random variable representing the occupancy
   - Cell is occupied: $p(m_i) = 1$, not occupied: $p(m_i) = 0$, no knowledge: $p(m_i) = 0.5$
   - The state is assumed to be static
2. **The cells (= random variables) are independent of each other**
3. **We have a sensor model that can measure free and occupied space**
4. **The likelihood $l_{t,i}$ of a cell being free can be represented by**

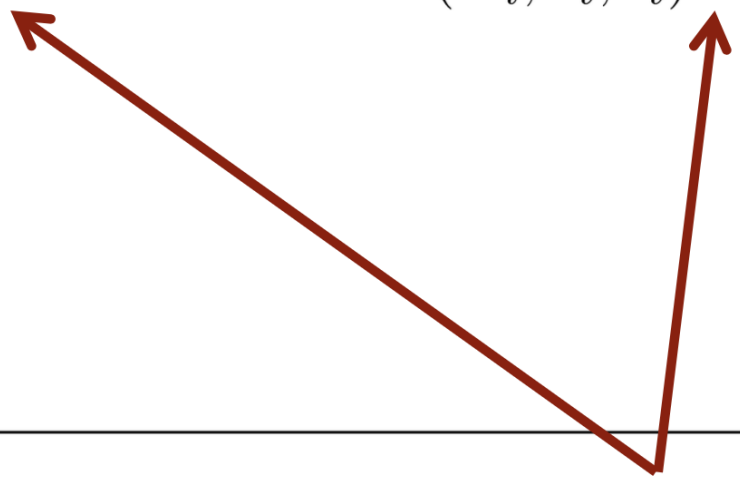
$$l(m_i|z_{1:t}, x_{1:t}) = \underbrace{l(m_i|z_t, x_t)}_{\text{Inverse sensor model}} + \underbrace{l(m_i|z_{1:t-1}, x_{1:t-1})}_{\text{recursive term}} - \underbrace{l(m_i)}_{\text{prior}}$$

or in short $$l_{t,i} = \text{inv_sensor_model}(m_i, x_t, z_t) + l_{t-1,i} - l_0$$

Occupancy Mapping Algorithm

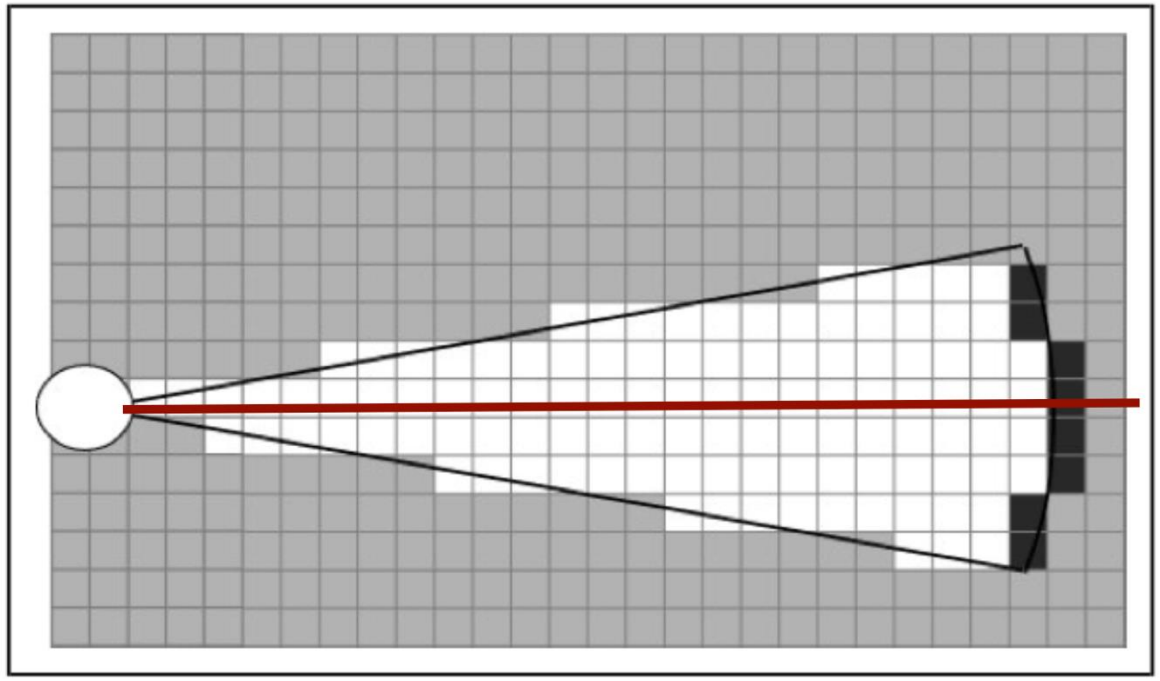
occupancy_grid_mapping$(\{l_{t-1,i}\}, x_t, z_t)$**:**

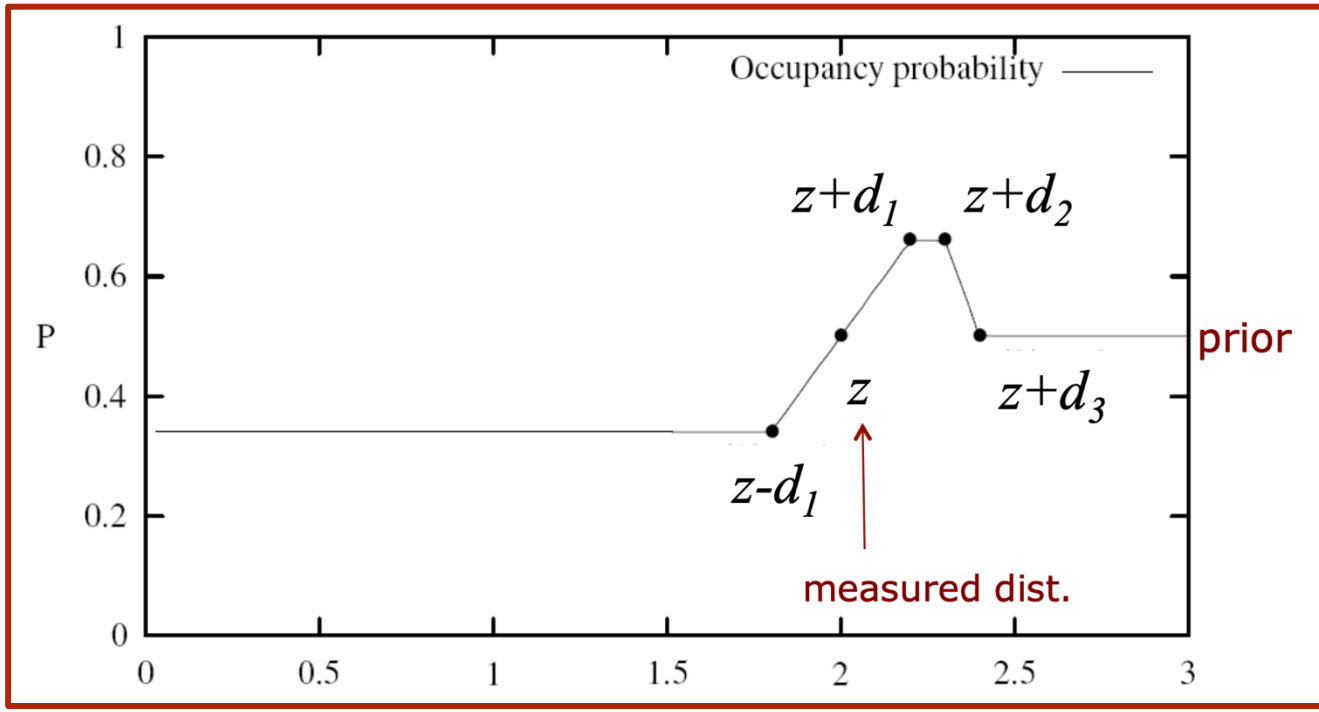
1: *for all cells* m_i *do*
2: *if* m_i *in perceptual field of* z_t *then*
3: $l_{t,i} = l_{t-1,i} + \text{inv_sensor_model}(m_i, x_t, z_t) - l_0$
4: *else*
5: $l_{t,i} = l_{t-1,i}$
6: *endif*
7: *endfor*
8: *return* $\{l_{t,i}\}$



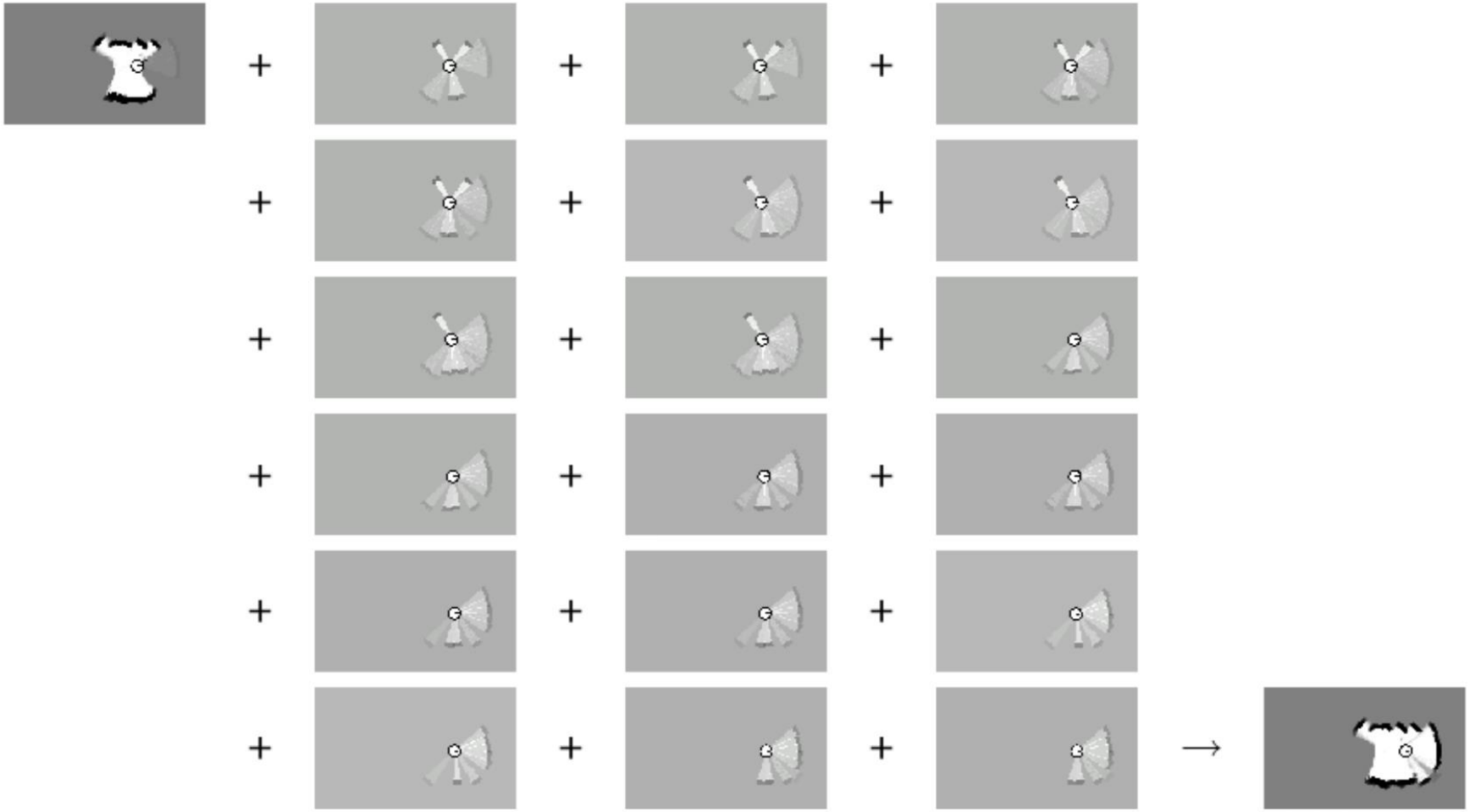
highly efficient, only requires to compute sums

Inverse Sensor Model for Sonar Range Sensors

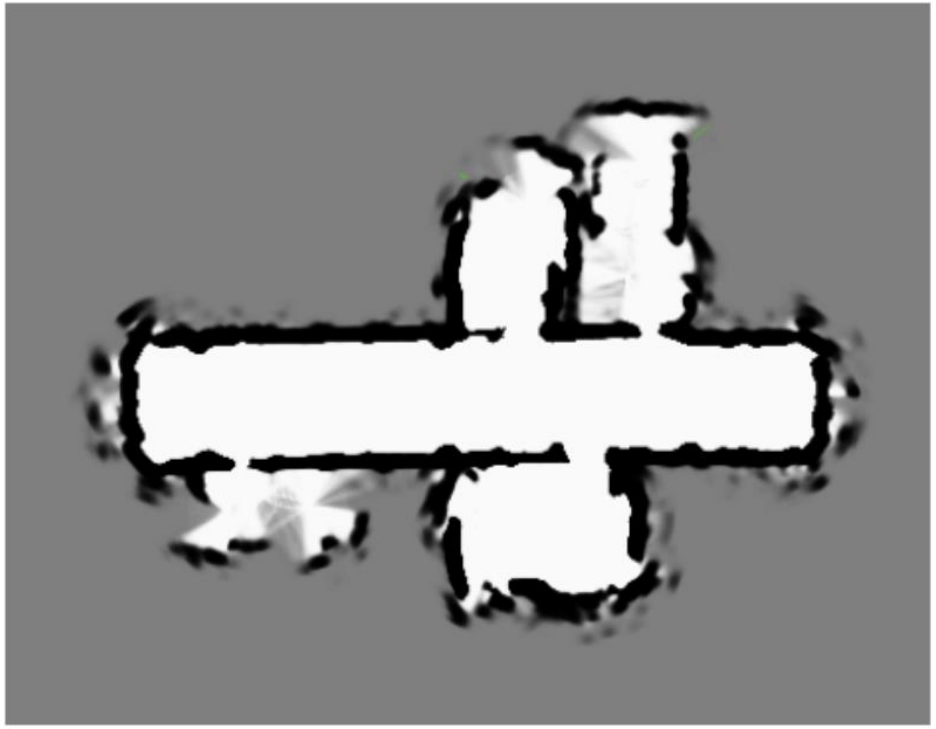


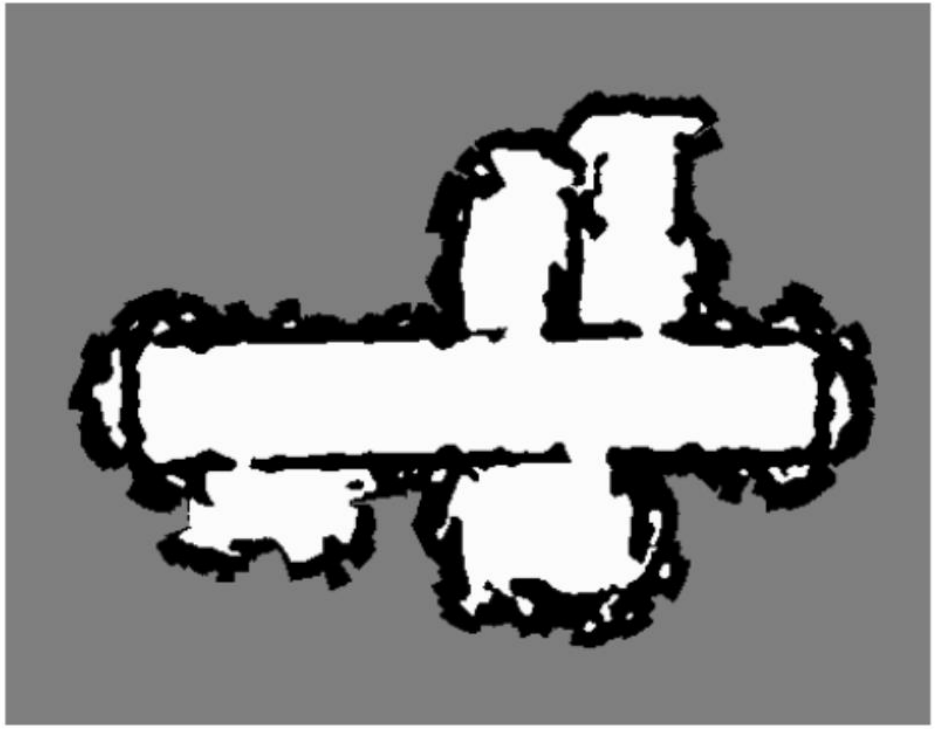


Example: Incremental Update on Occupancy Grids



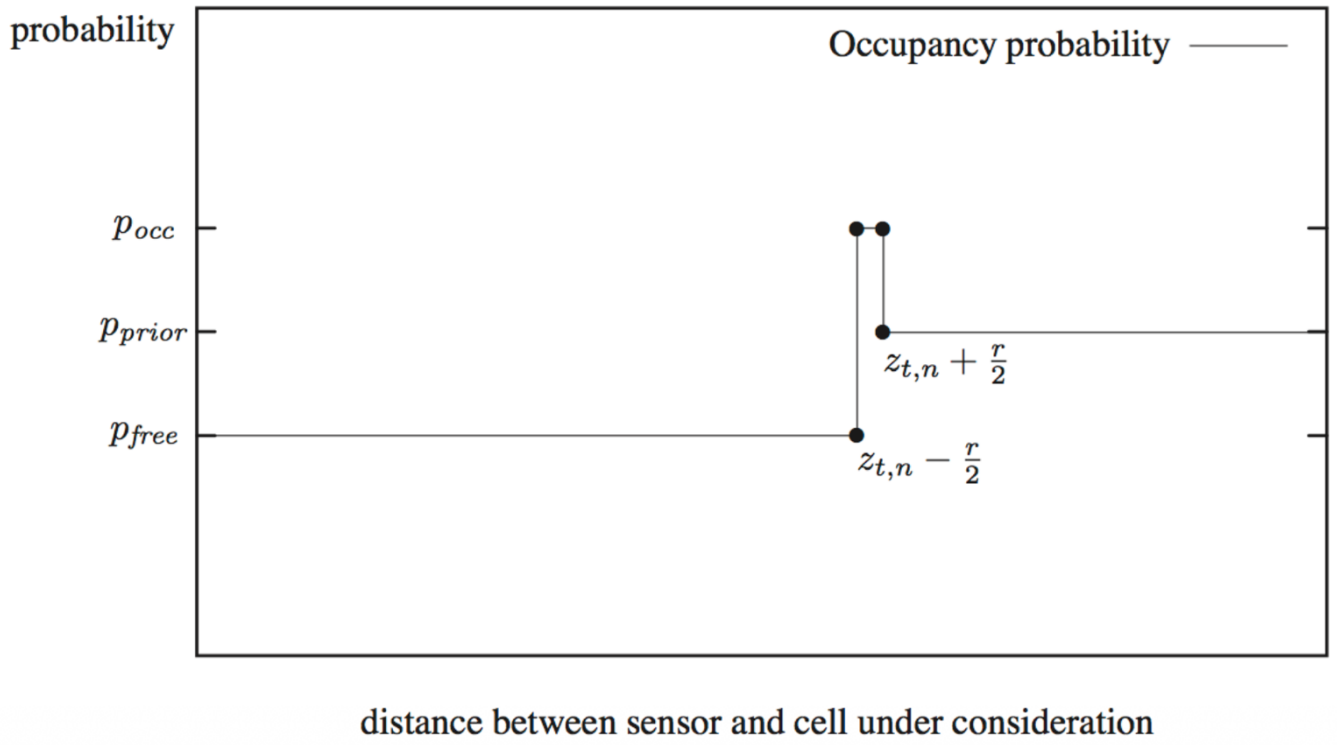
Resulting Occupancy Grid and Maximum Likelihood Map



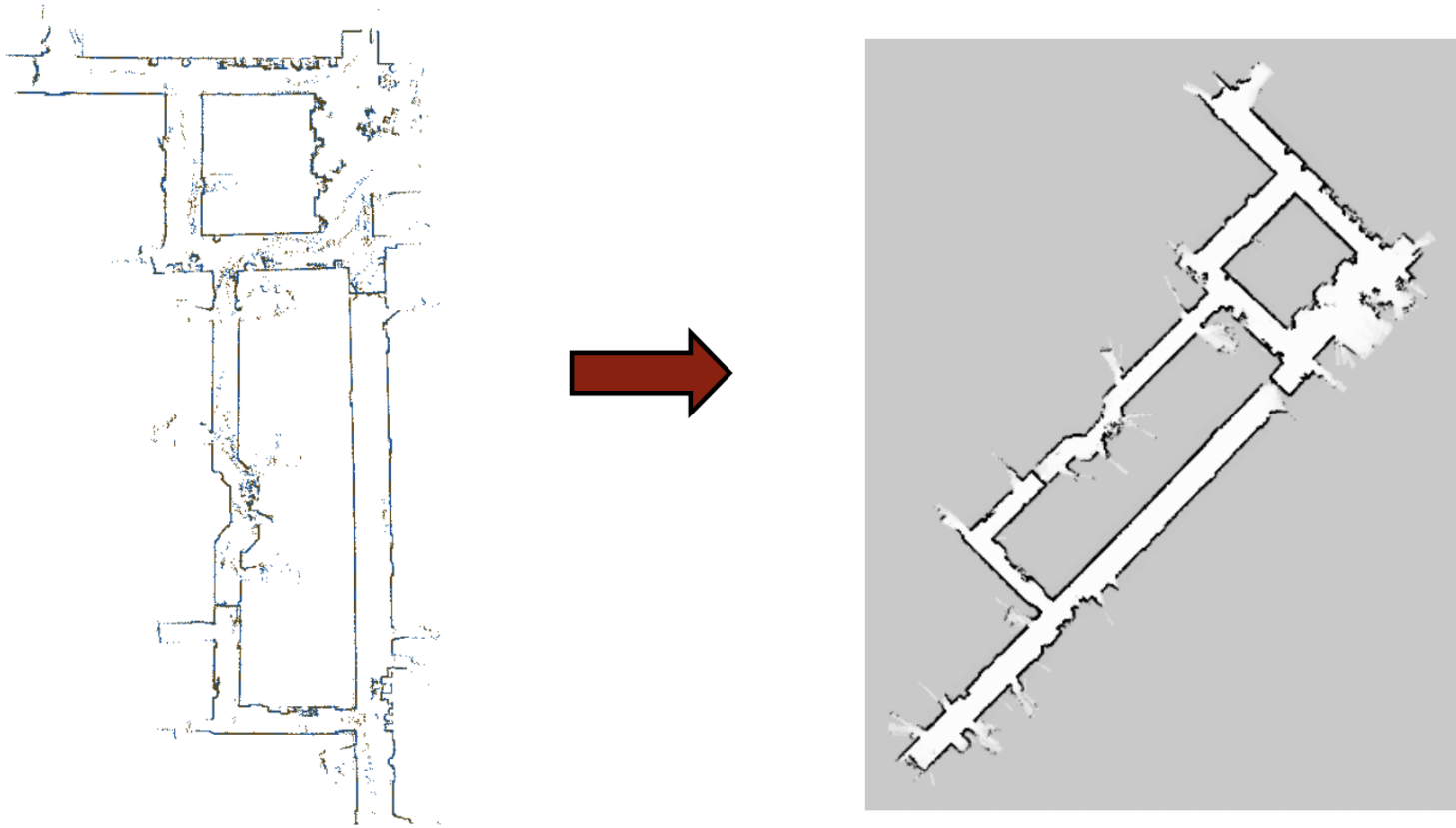


- **The maximum likelihood map is obtained by rounding the probability for each cell to 0 or 1.**

Inverse Sensor Model for Laser Range Finders



From Laser Scans to Maps

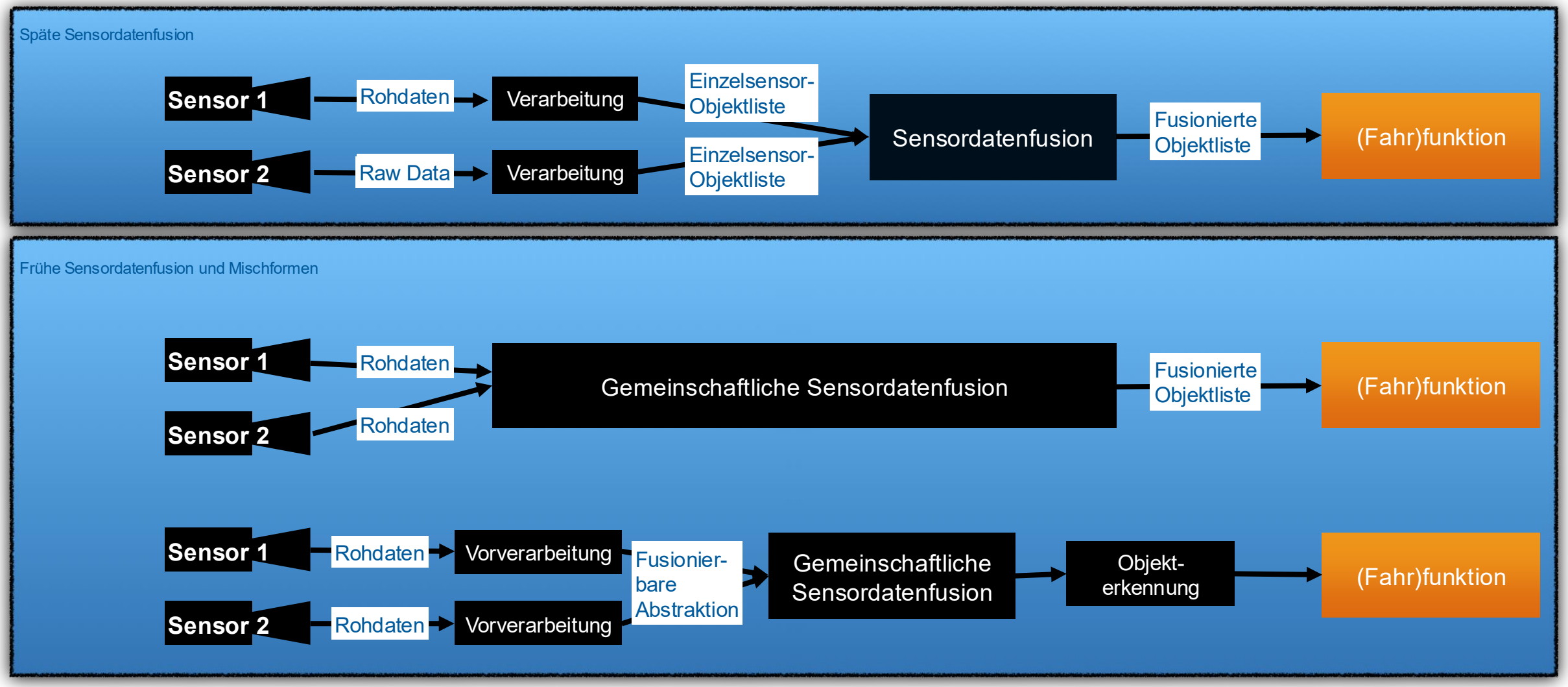


Summary

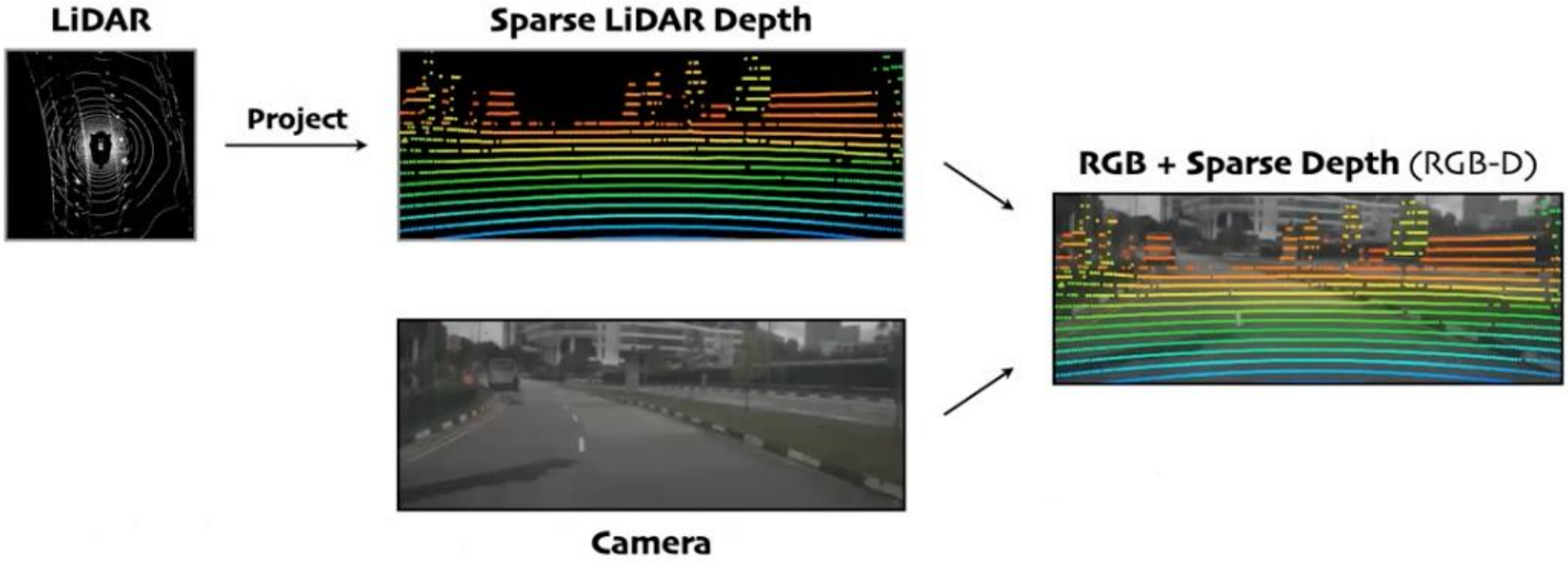
- Occupancy grid maps discretize space into independent cells
- Each cell is a binary random variable estimating if the cell is occupied
- Static state binary Bayes filter per cell
- Mapping with known poses is easy, cell states are fast to compute
- No need for predefined features
- However, only solves the mapping part ("known poses"), not the localization part.

Early Fusion

Overview

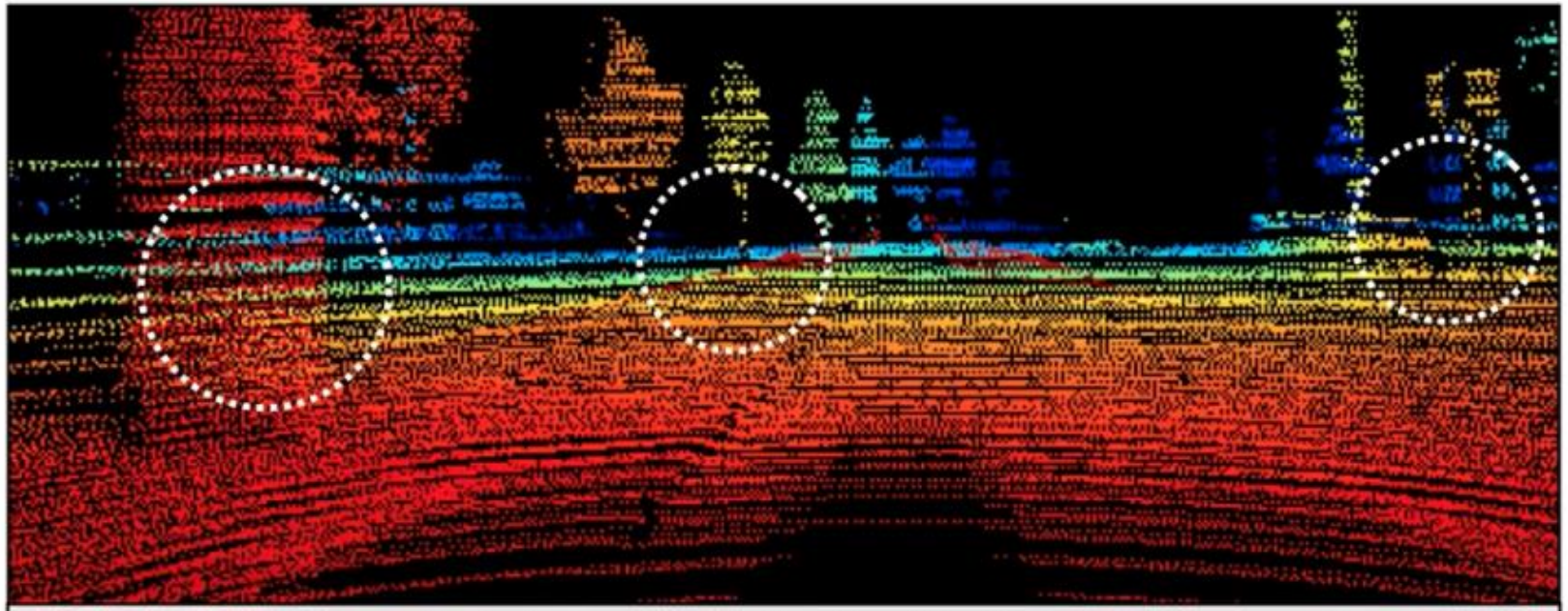

Späte Sensordatenfusion
Sensor 1
Rohdaten
Verarbeitung
Einzelsensor-Objektliste
Sensor 2
Raw Data
Verarbeitung
Einzelsensor-Objektliste
Sensordatenfusion
Fusionierte Objektliste
(Fahr)funktion
Frühe Sensordatenfusion und Mischformen
Sensor 1
Rohdaten
Sensor 2
Rohdaten
Gemeinschaftliche Sensordatenfusion
Fusionierte Objektliste
(Fahr)funktion
Sensor 1
Rohdaten
Vorverarbeitung
Sensor 2
Rohdaten
Vorverarbeitung
Fusionier-bare Abstraktion
Gemeinschaftliche Sensordatenfusion
Objekt-erkennung
(Fahr)funktion

Solution 1: Camera-Centric Model



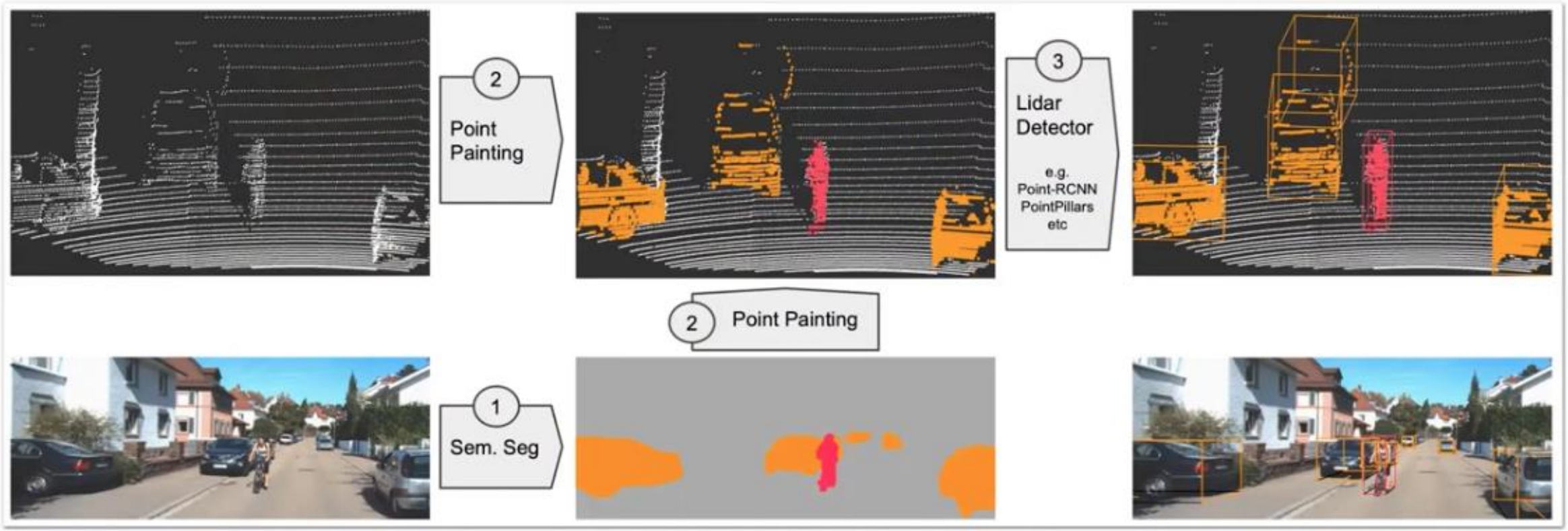
Images: Zhijian Liu, **Efficient 3D Perception for Autonomous Vehicles**

2D Projection Loses Distance Information

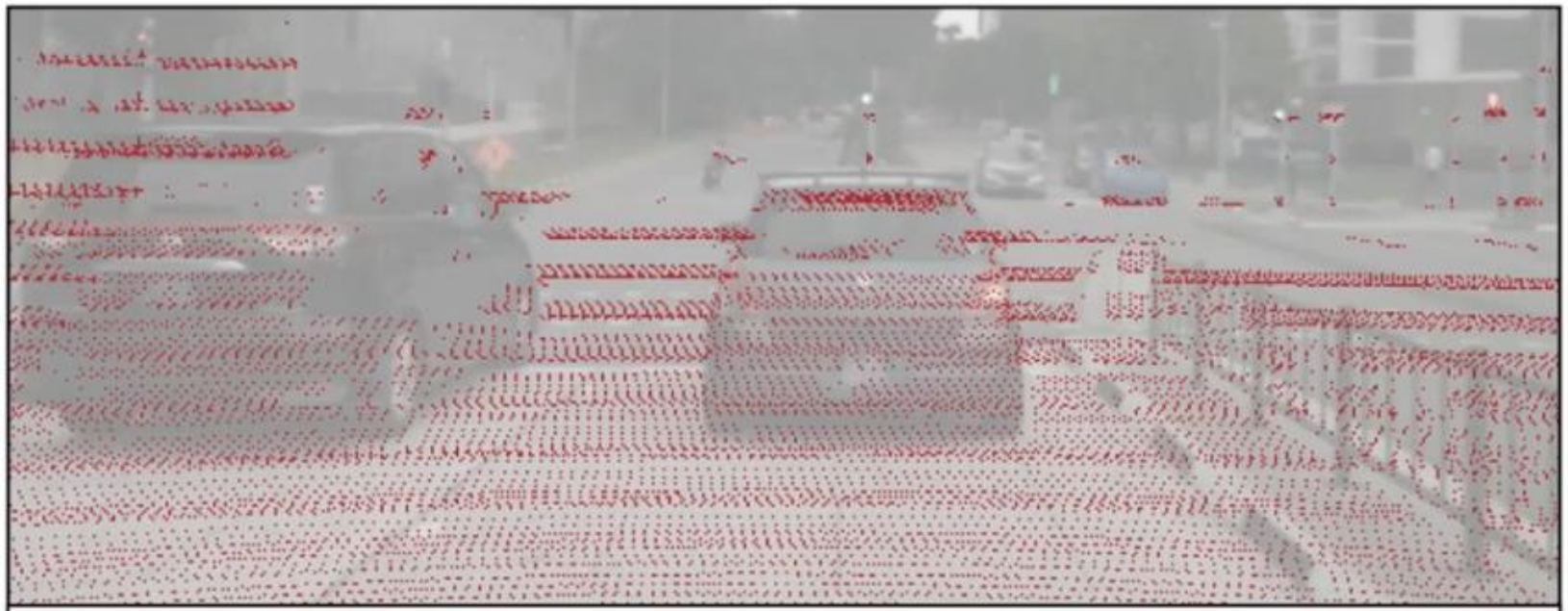


Neighbors in the projected **2D space** (after the LiDAR-to-camera transformation) could be **far away** from each other in the **3D space**.

Solution II: Lidar-Centric Fusion

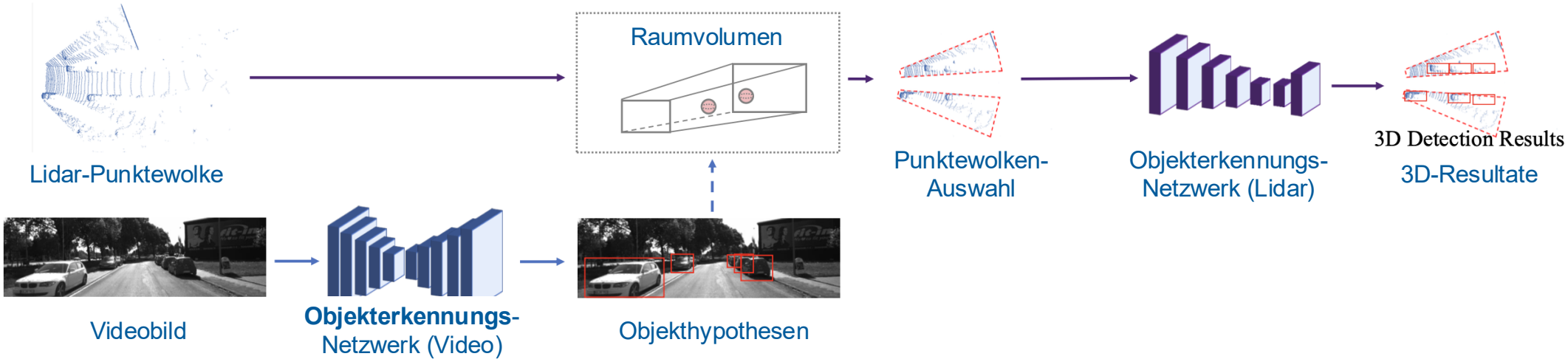


Only a small subset of camera features can be matched to lidar features ➔ semantically lossy

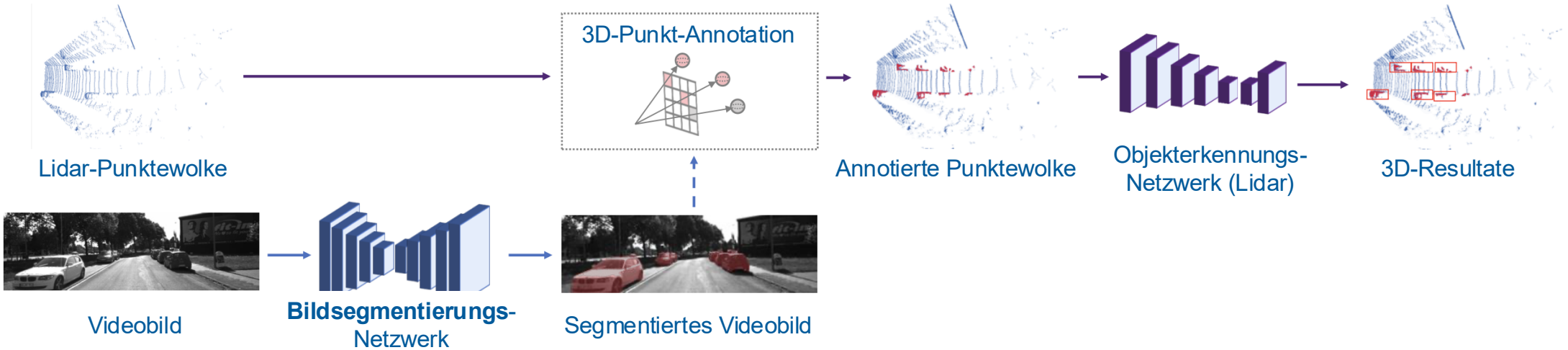


Only **5%** of camera features are **matched** and **decorated** to LiDAR points (for 32-beam LiDAR scanner).

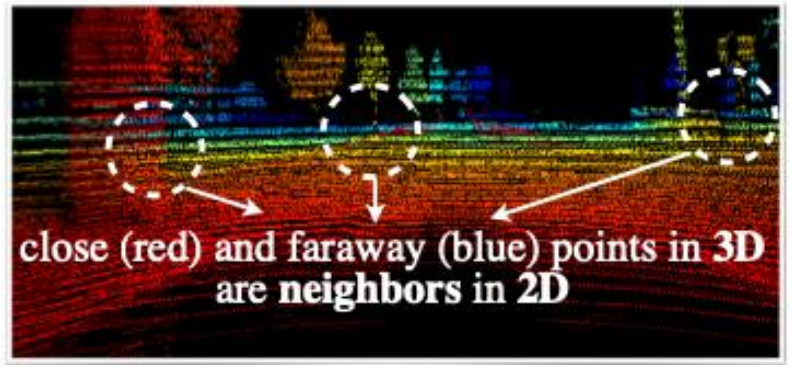
Sensor Networks and Sensor Fusion



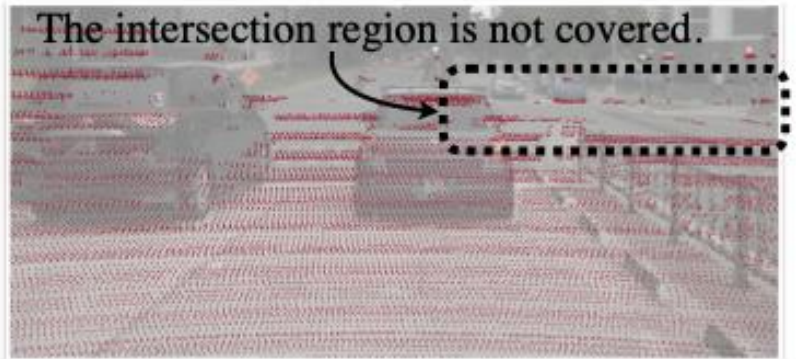
Fusion auf Regionsbasis



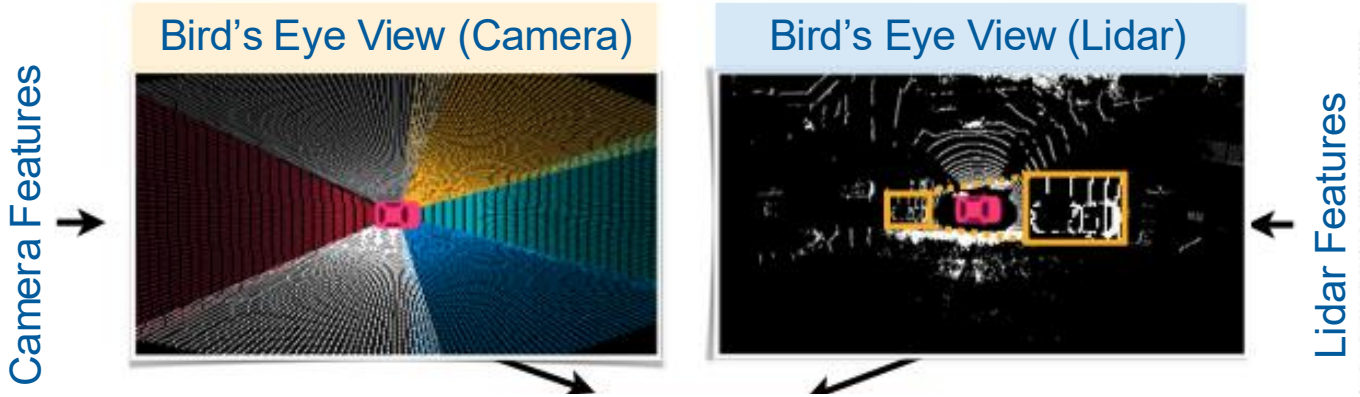
Fusion auf Punktebasis



Projecting lidar data into camera
loses 3D information

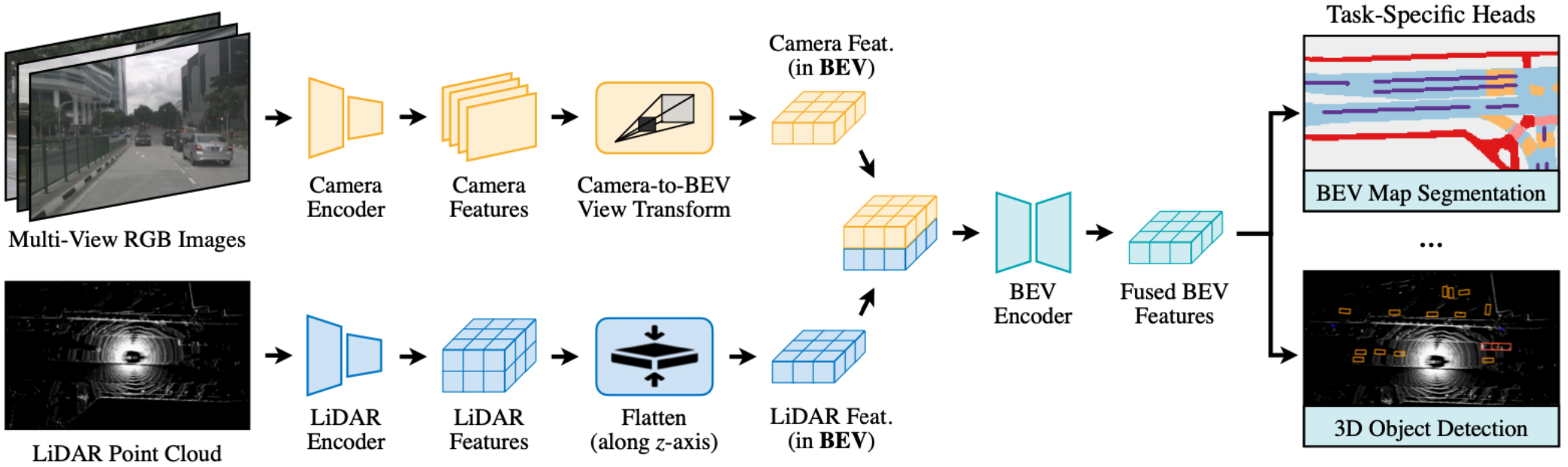


Projecting camera data into lidar
loses data density



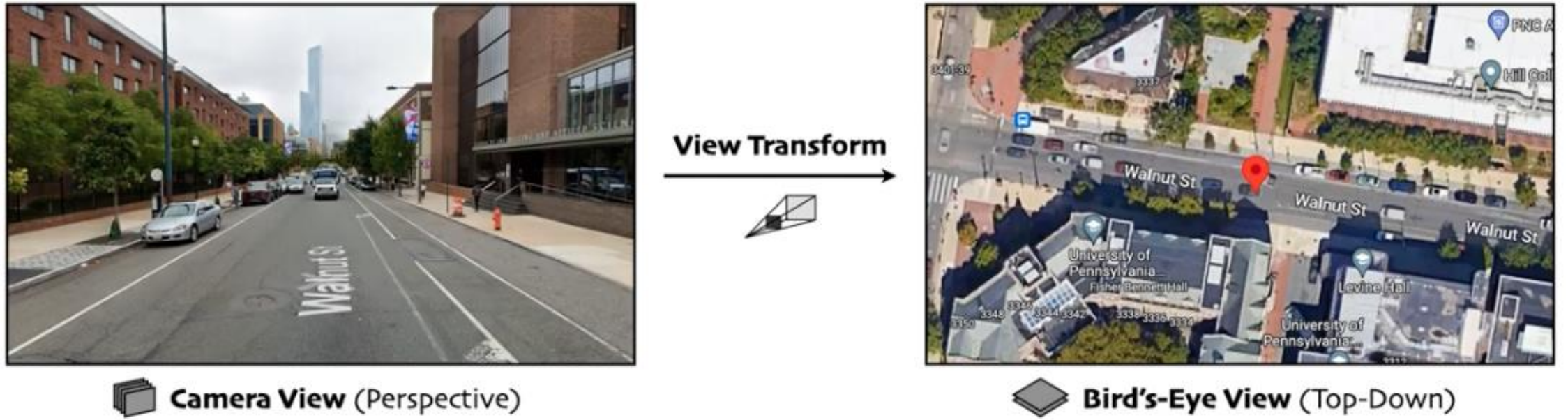
Common Space: Bird's Eye View

Camera Lidar Fusion in Bird's Eye View



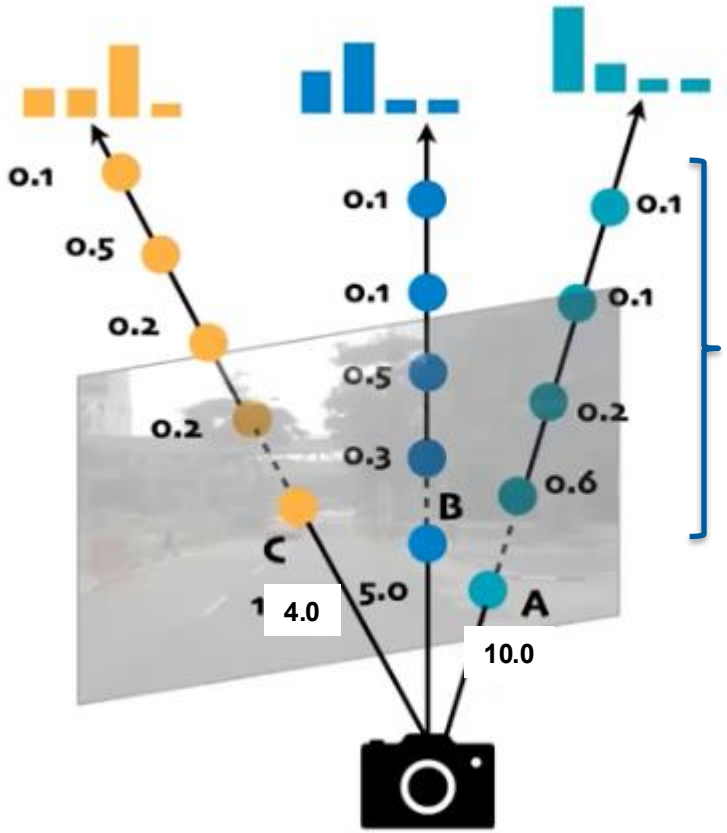
Camera to Bird's Eye View Transform

General Idea:



Each point in a camera image can have a feature value for belonging to a certain class (car, ped, lane marking, ...) and a distance / depth value along the ray going from the optical center to the point.

Camera to Bird's Eye Fusion: Step 1 – Depth Estimation



n classifier output nodes spaced at equidistant depth intervals

Depth Distribution

A	0.6	0.2	0.1	0.1
B	0.3	0.5	0.1	0.1
C	0.2	0.2	0.5	0.1

Image Features

A	10.0
B	5.0
C	4.0

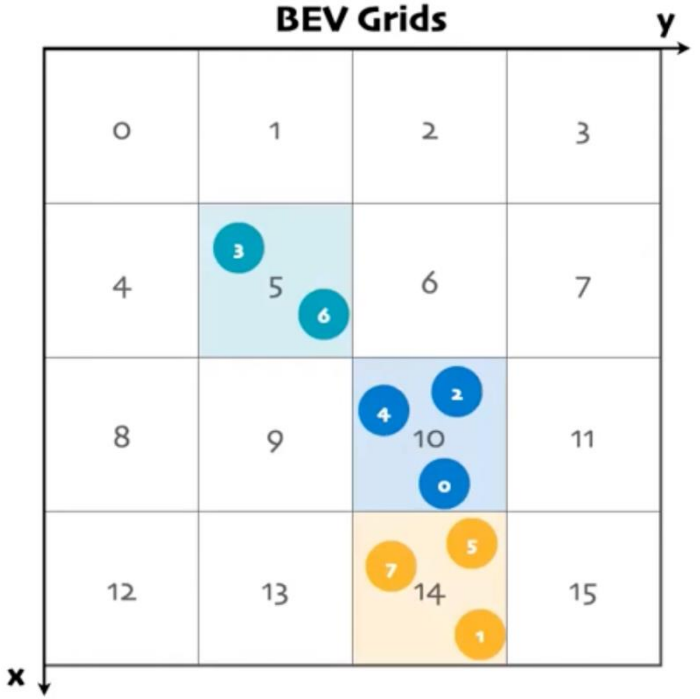
→

A	6.0	2.0	1.0	1.0
B	1.5	2.5	0.5	0.5
C	0.8	0.8	2.0	0.4

Point Features

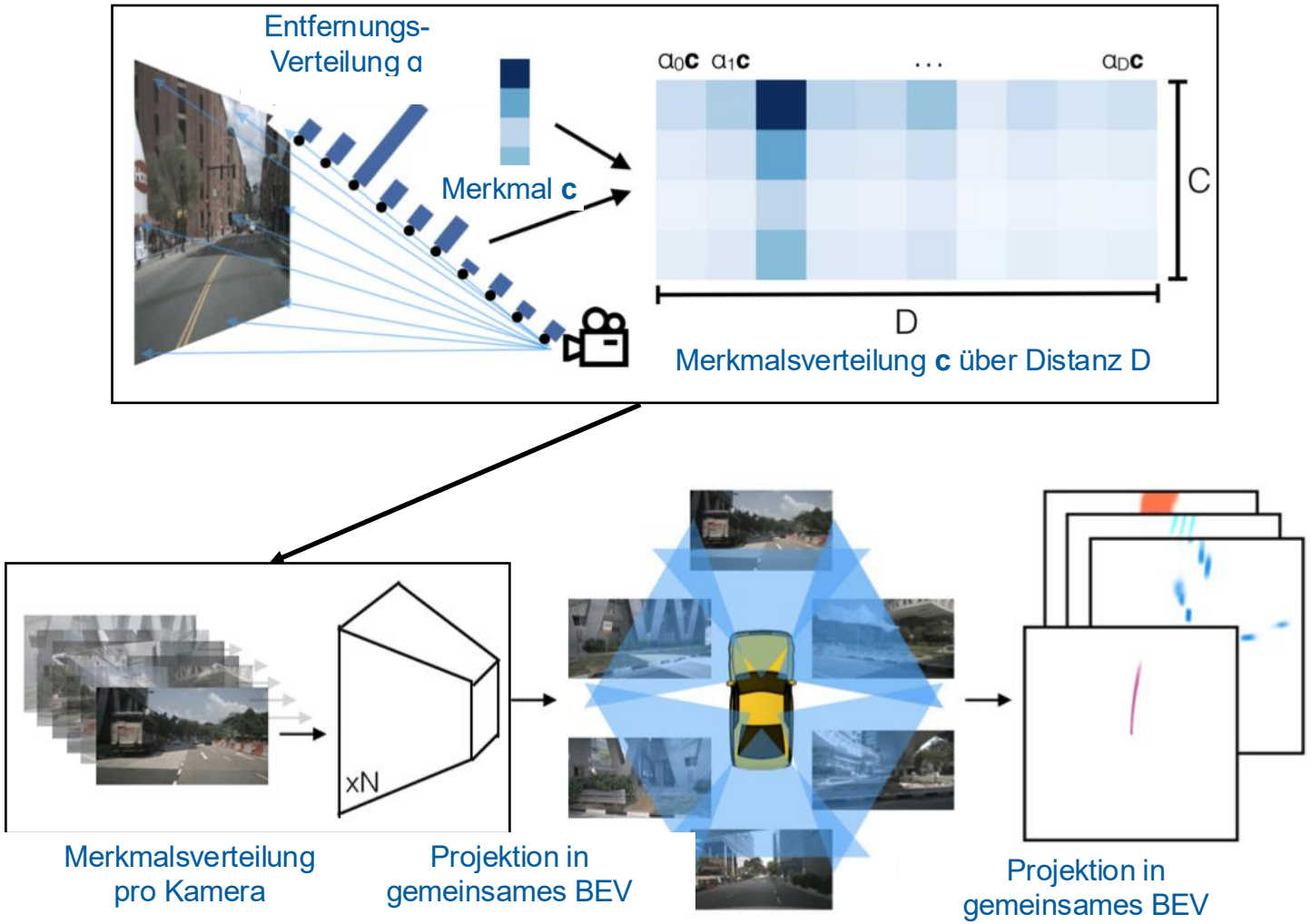
Camera to Bird's Eye Fusion – Step 2: Map features to grid and aggregate

Point			BEV Grid
Index	Coordinate	Feature	Index
0	(2.8, 2.6)	-2	10
1	(3.8, 2.9)	6	14
2	(2.1, 2.7)	7	10
3	(1.2, 1.3)	1	5
4	(2.3, 2.1)	-1	10
5	(3.1, 2.8)	4	14
6	(1.7, 1.8)	3	5
7	(3.4, 2.2)	-3	14

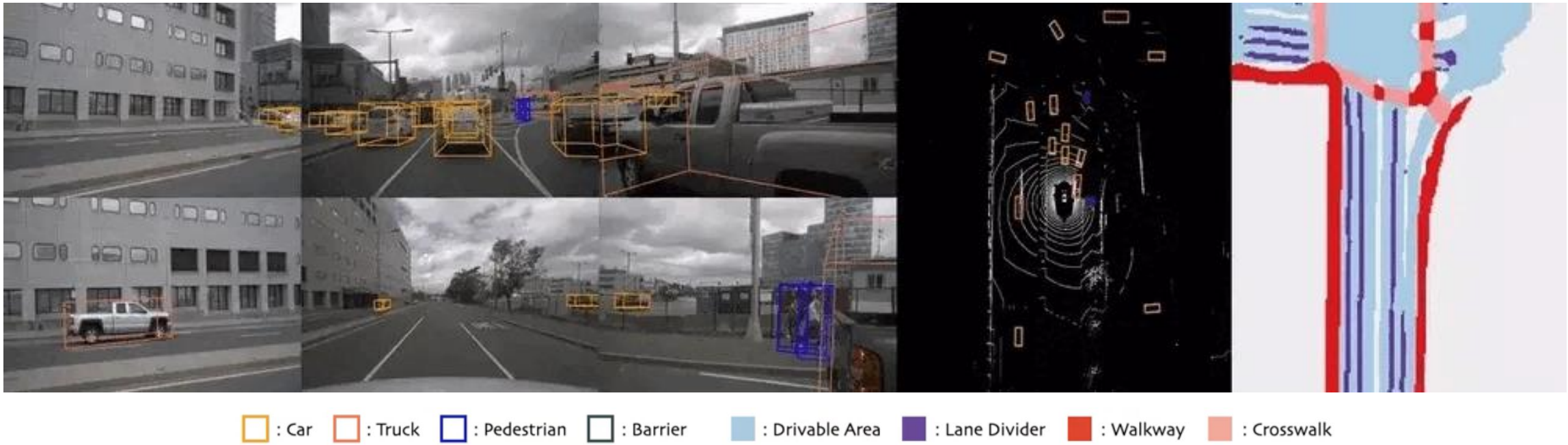


Grid Cell	Feature Index	Feature Value	Sum
5	3	1	4
	6	3	
10	0	-2	4
	2	7	
	4	-1	
14	1	6	7
	5	4	
	7	-3	

BEV Fusion – Example for Early / Mid Fusion



Demo


: Car
: Truck
: Pedestrian
: Barrier
: Drivable Area
: Lane Divider
: Walkway
: Crosswalk

Literature / Links



1. **Alberto Elfes: *Using Occupancy Grids for Mobile Robot Perception and Navigation,* https://www.cs.cmu.edu/~motionplanning/papers/sbp_papers/integrated4/elfes_occup_grids.pdf**



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Thank you very much!